Myungsin Cho

Teaching Statement

I have taught mathematics at different levels and in a range of formats. My experience spans general-education courses for non-majors, service courses such as calculus, proof-based linear algebra, and applied mathematics for STEM students. I have taught in traditional classrooms and large lectures with more than 100 students, as well as in flipped classrooms and synchronous online lectures.

Across these settings, I want students to experience mathematics as one connected story rather than a collection of facts and procedures. I organize courses around questions that motivate new concepts and help students see how mathematical ideas develop from one another. My goal is for students to master fundamental skills while learning to explain their reasoning, evaluate arguments, and pursue mathematical questions for themselves.

Connecting Questions and Concepts

A logically organized course does not always make its connections apparent to students. I therefore consider when and how to introduce ideas so that their motivations become clear. I return to earlier questions as new concepts become available, allowing students to recognize how different topics fit together.

In Calculus I, I organized limits, derivatives, and integrals around approximation as a unifying theme. I used a similar approach in *Analysis and Optimization*, where linear approximation and the gradient help identify candidates for extrema, while quadratic approximation and the Hessian help classify their local behavior. The limitations of first-order information motivate the need for second-order methods, and gradient descent shows how local information can also guide numerical computation.

In proof-based Linear Algebra, I structured the course around questions that students could revisit as they developed a deeper understanding. We began with matrices and systems of linear equations, learning concrete methods of computation while leaving some fundamental questions unresolved. As students encountered vector spaces and linear maps, we returned to these questions from a new perspective. For example, the definition of matrix multiplication became natural through the composition of linear maps. In this way, abstract concepts arose as answers to questions students had already encountered.

I also emphasize patterns of reasoning that recur across different subjects. Questions of existence, uniqueness, and classification help students formulate mathematical problems, even when the objects under study change. When reading a proof, I ask students to identify its central idea and consider how the author might have discovered the argument. Recognizing these patterns helps students approach unfamiliar problems rather than simply reproduce arguments they have seen.

Supporting Students' Learning

I adapt my teaching to the purpose of each course and to the students taking it. As a instructor for Math of Beauty and Decision, a general-education course for non-majors, I used familiar examples involving voting, symmetry, and games to introduce mathematical ideas. In courses for STEM students, I emphasize the mathematical foundations and reasoning skills they will need for further study, including understanding when and why to apply particular methods.

Students within the same course also have different levels of preparation and mathematical maturity. In *Analysis and Optimization*, students come from various STEM fields, and many have little experience writing formal proofs. When I noticed that students struggled to begin proofs, I redesigned some homework assignments as guided worksheets. These provide the structure of an argument while asking students to develop and justify key steps themselves. Students reported that the worksheets made proofs more accessible and intuitive.

Students' questions and written work help me adjust my teaching throughout the semester. During class, I invite students to anticipate the next step or explain why an argument should work. In office hours, I ask where their reasoning stopped being clear and work from that point, often using a concrete example or a different explanation. I also enjoy learning about students' academic interests and advising them on future coursework and graduate study. I want students to feel comfortable asking questions and contributing their ideas, regardless of their mathematical preparation.

I design assessments to make regular practice part of learning while maintaining clear expectations for mathematical reasoning. In my current *Analysis and Optimization* course, homework provides low-stakes practice, weekly quizzes encourage consistent preparation, and exams assess computation, conceptual understanding, and proof. Quiz questions are selected from assigned homework problems, and quiz performance can count toward the routine-computation portion of the corresponding exam. This rewards mastery of fundamental skills through regular practice while allowing exams to devote greater attention to concepts and proofs.

I also use grading policies that recognize students' progress over time. In previous courses, I have given greater weight to students' stronger exam performances and allowed a final examination to replace a lower earlier exam score. In Calculus, I allowed students to correct missed exam problems for partial credit. These approaches encourage students to revisit their mistakes and demonstrate improvement rather than treating an early setback as definitive. I continue to reconsider how the timing, weighting, and structure of assessments can better support learning.

Developing Independent Mathematical Thinking

Communicating mathematical reasoning is an important step toward independent thinking. In Linear Algebra, I used an oral final examination and provided sample problems and possible follow-up questions in advance. Preparing their explanations gave students an opportunity to identify gaps in their understanding. During the examination, I could ask them to clarify a step or explain their choice of method, allowing me to assess their reasoning beyond the correctness of a written answer.

The growing use of AI makes independent mathematical judgment increasingly important. I emphasize to students that using AI properly and responsibly requires more than writing effective prompts. They need enough mathematical understanding to examine assumptions, identify gaps in reasoning, and verify conclusions produced by these tools. I generally require handwritten homework, giving students another opportunity to work through and check their reasoning before submitting it, even when they have consulted external tools.

My experience mentoring undergraduates through a Directed Reading Program has also shaped my approach to independent mathematical inquiry. I encouraged students to raise their own questions and explain the material they studied, rather than simply complete assigned readings and exercises. In undergraduate research, I would build on this experience by beginning with accessible mathematical examples and helping students turn their observations into questions they can investigate. Examples and calculations can lead to conjectures, proofs, or counterexamples, with students taking greater responsibility for deciding what to explore next. I believe that learning to formulate and pursue mathematical questions is itself a valuable research experience.

I will continue to refine my teaching through students' questions, their work, and their feedback. I look forward to bringing this approach to a broad range of undergraduate mathematics courses and to mentoring students as they develop their own mathematical interests.